How Moirai-2.0 makes a forecast

Salesforce · 11M parameters (small) · decoder-only transformer · zero-shot

1. One series at a time

Moirai 2.0 reads only the target's own past. A ? is a value not known yet.

past *future*

Target temperature

It was trained on single series, so covariates are not passed.

Five metal prices become five separate forecasts.

2. Scale, then patches of 16

The context is scaled by its mean μ and standard deviation σ. Each patch keeps a flag for missing values.

$$\tilde{x} = \frac{x - \mu}{\sigma}$$

value

missing?

one patch = 16 steps

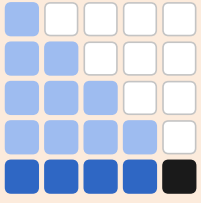
residual MLP

one token: 384 numbers

3. Six causal blocks

Each patch reads only itself and earlier patches. RMSNorm in front, shortcut around.

earlier → later



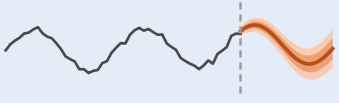
earlier patches only

Self-attention: causal, 6 heads, rotary positions (RoPE)

Feed forward: SwiGLU, 384 → 1,024 → 384

4. Four patches per token

The last token gives **9 quantiles** for each step of the next **4 patches (64 steps)**.

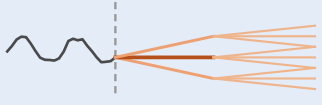


median

60% band

80% band

Longer horizons: all 9 quantile paths go back in, and 9 quantiles are taken from the 81 results.



9 paths × 9 = 81 → 9 quantiles

A small decoder trained on quantiles that keeps the spread when it rolls forward.

6 blocks · width 384 · 6 heads · patches of 16 · 4 patches × 9 quantiles per token

In the notebooks: last 1,680 days or full monthly history · no covariates · median as the point · 90 days = 2 passes · CC-BY-NC-4.0